

SETU

The Cognitive Operating System — Bridging the Gap Between Dense Digital Worlds and the Neurodivergent Mind

Theme: Disability Inclusion & Accessibility**Target Audience:** ADHD, Dyslexia, Autism, Memory Disorders**Hackathon:** Capgemini Hack4Positive 2026

1. EXECUTIVE SUMMARY & CORE PARADIGM SHIFT

Most modern assistive applications are built as fragmented, single-deficit tools (e.g., standalone screen readers or basic text highlighters). Real-world digital navigation is significantly more multi-dimensional and overwhelming. **SETU** (meaning "Bridge") completely reimagines cognitive accessibility by launching an unified, cross-surface **Cognitive Operating System (COS)**. It places an intelligent, adaptive orchestration layer between a person with cognitive variations and dense digital interfaces.

By integrating an instantaneous, client-side browser extension with a deep-work web workspace and an advanced mobile companion, SETU seamlessly transforms complex websites, dense academic PDFs, overwhelming data structures, and chaotic workflows into predictable, low-cognitive-load execution paths. The underlying framework relies on a unified, shared AI engine optimized for structured, predictable JSON schemas rather than unpredictable chat interfaces, maximizing reliability, speed, and trust.

The Trillion-Dollar Ideator Perspective: Products win when they eliminate the friction of *task initiation*. For neurodivergent populations, the "Wall of Awful" is the primary barrier to execution. SETU transitions accessibility tools from passive readers to active, predictive partners.

2. PRODUCT ECOSYSTEM & SURFACE ARCHITECTURE

SETU is designed as a single, fully synchronized application architecture delivered through three specialized interactive surfaces:

A. The SETU Lens (Chromium Browser Extension)

Acts as the real-time interaction and DOM-modification layer. It intercepts web pages natively, re-rendering them locally to match individual neurological comfort thresholds without introducing network latency or backend computational costs.

B. The SETU Sanctuary (Knowledge Base Web Application)

The dedicated personal command center for high-retention deep work. Accessible directly via a native one-click integration built into the extension, it enables users to seamlessly ingest personal files (PDFs, complex research papers, dynamic URLs, or captured images). The shared AI orchestration engine processes this information, stripping away distracting semantics and re-rendering the core knowledge as interactive visual layouts.

C. The SETU Go (Mobile Companion & Browser Sync)

An advanced Progressive Web App (PWA) / native Android hybrid layer that syncs state seamlessly with the web workspace. It enables mobile web browsers to execute the exact same core UI modifications (Bionic Reading, Line

Highlighting) while providing on-the-move access to the Knowledge Base, ensuring cognitive continuity across all devices.

Surface Entry Point	Core Accessibility Objective
SETU Lens (Browser Extension Overlay)	Real-time web transformation, clutter elimination, local processing, and voice-driven DOM navigation.
SETU Sanctuary (Web Knowledge Base)	Deep processing of complex materials into visual maps, automated summaries, and persistent memory retention tools.

3. CORE FEATURES BLUEPRINT

The primary feature set addresses explicit cognitive processing bottlenecks, ensuring immediate usability and a striking visual presentation during live hackathon evaluations.

- **Bionic Reading & Line Highlighting:** Dynamically alters typography by bolding the initial fixations of words and adding a high-contrast horizontal line guide. This anchors visual attention, bypassing phonological loop bottlenecks common in Dyslexia and drastically enhancing reading velocity and comprehension.
- **Sensory-Safe Focus Mode:** A structural DOM sanitizer that extracts core semantic components and strips away tracking code, advertisements, blinking visual components, auto-playing multi-media elements, and chaotic CSS grids, rendering a clean, distraction-free layout.
- **SETU Commander (Voice-Driven Autonomous Web Agent):** Bypasses complex spatial coordination barriers. Users give natural voice instructions (e.g., *"Fill this contact form with my profile details"* or *"Navigate to the checkout button"*), and the local agent compiles the intent, parses the underlying DOM tree, maps fields, and executes the multi-step browser interactions seamlessly.
- **Multilingual Vernacular Explainer:** When executing voice-based inquiries regarding technical jargon, dense academic frameworks, or complex legal parameters, the shared AI engine translates, adapts, and speaks the explanation in the user's explicit or native language, using clear real-world analogies tailored to their exact preference.
- **Complex UI & Image Breakdown:** Allows users to instantly isolate intricate charts, multi-axis infographics, or dense diagrams to generate descriptive textual breakdowns or sequential step-by-step plain-language cards.
- **Cognitive Ingestion & Interactive Mapping:** Within the Knowledge Base, any uploaded article, document, or textbook is synthesized by the shared AI engine and instantly converted into interactive, expand-and-collapse **Mind Maps** and structural flowcharts, avoiding the visual fatigue of continuous text blocks.

4. ADVANCED "WINNING EDGE" FEATURES (HACKATHON DIFFERENTIATORS)

To firmly establish SETU as the undisputed gold-standard winner of the Capgemini Hack4Positive 2026 Hackathon, we introduce three pioneering, high-impact features engineered to captivate engineering judges and product specialists alike:

UNICORN INNOVATION 1 The "Breathe" Protocol (Behavioral Overwhelm Detection)

SETU tracks client-side interaction signals like erratic cursor movements, rapid chaotic scrolling patterns, or repetitive "rage-clicking" on complex web structures. Upon detection, SETU captures this marker of cognitive overload or executive dysfunction, smoothly dimming the interface to overlay a 5-second animated box-breathing visual prompt. Once normalized, it gently asks: *"This page looks a bit intense. Shall I simplify it and highlight the next step for you?"*

UNICORN INNOVATION 2 The Contextual Memory Vault (Local Vector RAG)

Specifically engineered to support individuals suffering from short-term memory disorders, dementia, or severe brain fog. SETU runs a localized vector database (pgvector backend) that indexes historical contexts, read documents, and user research trails. If a user returns to a multi-day research process and suffers a memory lapse, they can simply ask the voice companion: *"What was I learning about yesterday afternoon?"* SETU instantly reconstructs a pristine, chronological mental map and progress card.

UNICORN INNOVATION 3 **Preemptive Cognitive Task Chunking**

When navigating to structurally dense, high-anxiety digital portals (such as public banking systems, employment portals, or government documentation registries), SETU preemptively intercept the architecture. Instead of presenting the terrifying full screen, it extracts requirements and populates a minimalist, gamified 3-step linear checklist over the page, allowing users to conquer severe task initiation paralysis effortlessly.

5. TECHNICAL ARCHITECTURE & HYBRID ORCHESTRATION

The system relies on a high-throughput, low-latency stack designed to run complex processing pipelines on the client side while utilizing cloud orchestration where heavy intelligence is required.

Layer	Technology Recommendation	Architectural Purpose & Advantage
Frontend Core & Ext	Next.js 15, TypeScript, Tailwind CSS, Manifest V3	Guarantees blazing-fast rendering speeds, strict type-safety, and secure browser extension side-panel hooks.
Local Execution	WebAssembly (WASM), Rust-compiled parsers	Executes high-frequency text-munging and Bionic Reading DOM rewrites purely locally within the user's browser runtime.
AI Orchestration	Vercel AI SDK, OpenAI GPT-4o / Gemini 1.5 Pro	Dynamic multi-model fallback routing. Enforces rigid JSON schemas to output deterministic data structure models for UI rendering.
Persistence & Vector	Supabase (PostgreSQL) + pgvector	Maintains synchronized user preference DNA profiles, cached document embeddings, and the Contextual Memory Vault.
Audio & Voice	Whisper API + Edge-driven TTS Functions	Enables real-time, low-latency, localized voice execution commands and fluid multilingual vernacular text-to-speech feedback.

6. STRATEGIC SCALE & ENTERPRISE-GRADE SECURITY

A great hackathon idea becomes a unicorn when it proves it can scale sustainably to millions of active users while protecting sensitive human datasets.

A. The "Zero-Compute" Scaling Model

By moving structural DOM modifications, bionic reading, and focus layouts completely to **WebAssembly (WASM)** operating on the client's local CPU, the backend infrastructure footprint is fundamentally eliminated for core features. Millions of users can simultaneously run SETU's parsing capabilities without creating cloud database or server computing overhead. High-overhead AI processes (like document summarization and map generation) use cached vector lookups and edge computing nodes (Cloudflare Workers) to minimize operational latency and bandwidth costs.

B. Zero-Knowledge Compliance Architecture for PII

Handling data for vulnerable populations demands absolute privacy parameters. SETU implements a strict **Zero-Knowledge Encryption Standard**. Highly sensitive materials uploaded into the Knowledge Base (e.g., medical records, employment applications, personal notes) are fully encrypted client-side using localized AES-256 keys before synchronization to the persistent cloud database.

Furthermore, prompt engineering layouts enforce strict data privacy rules: all interactions processed via downstream foundational Large Language Models utilize ephemeral API instances governed by data-handling agreements that explicitly prohibit the ingestion of user payloads for base model training.

7. MENTOR PITCH STRATEGY & LIVE DEMONSTRATION FLOW

To maximize validation scores under the Hack4Positive evaluation rubric, the 5-minute live pitch must follow an emotional, high-conviction narrative arc:

1. **0:00 - 1:00 | Establish the Human Friction:** Introduce a student with severe ADHD and Dyslexia staring at a massive, complex 50-page university registration portal, experiencing visible cognitive freezing (the "Wall of Awful").
2. **1:00 - 2:15 | The Live Intervention:** Demonstrate the **SETU Lens** browser extension. Show the *Breathe Protocol* auto-activating as mouse erraticism increases, smoothing the page out instantly into a 3-step checklist with Bionic Reading enabled.
3. **2:15 - 3:30 | Deep Synthesis:** Click the "Send to Sanctuary" integration button. Show the immediate visual layout shift as the dense portal text blocks transform into a clean, interactive hierarchical Mind Map with memory flashcards generated on the side.
4. **3:30 - 4:30 | Prove Technical Superiority & Viability:** Present the Local-First WASM scaling metrics and the Client-Side Zero-Knowledge data encryption framework to dismantle any scaling, cost, or security doubts raised by technical judges.
5. **4:30 - 5:00 | The Grand Unicorn Vision:** Close with a memorable, high-impact statement: *"SETU does not simply read information aloud—it serves as the structural bridge that transforms overwhelming digital chaos into clear human action."*